

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Separate the Variables

1. Variable Separation

- a. The following is a method to find the *exact solution* to $y' = 0.2y$. For now assume that $y > 0$. This assumption corresponds to the population growth context and it will make the algebra easier and hence the underlying idea clearer.

Task	Show work here
Write y' as $\frac{dy}{dt}$. Then, divide both sides of $y' = 0.2y$ by y . —	_____
Replace $\frac{1}{y} \frac{dy}{dt}$ with $[\ln(y)]'$. —	_____
Write integrals with respect to t on both sides. —	_____
Apply the Fundamental Theorem of Calculus to integrate both sides. —	_____
Solve for y (and remember that y is actually a function, $y(t)$). —	_____
Show that the <i>general solution</i> can be written as $y(t) = C_1 e^{0.2t}$. —	_____

The end result, $y(t) = C_1 e^{0.2t}$ is called the *general solution* because it represents all possible functions that satisfy the ODE. The general solution is used to find any *specific solution*, which is a solution that corresponds to a given condition.

- b. Find the *specific solution* of the ODE $y' = 0.2y$ with initial condition $y(0) = 1$. Note that the general solution is $y(t) = C_1 e^{0.2t}$, which was shown in Part (a).

2. The Process of Separation of Variables (SOV)

The method of *Separation of Variables (SOV)* can solve a 1st-order ODE written in *separable form*,

$$y' = f(y)g(t)$$

where the expression $f(y)g(t)$ represents a function of y and t , which can be factored into two distinct functions: one with y and the other t .

We often circumvent explicit use of the chain rule and instead use a shortcut to more efficiently find the general solution. The shortcut involves treating the derivative y' as a ratio $\frac{dy}{dt}$, “separating” the dy and dt .

a. In the table below, follow the instructions to see how the shortcut works, using the equation $y' = -2ty$.

Task	Show work here
Determine $f(y)$ and $g(t)$. —	_____
<i>Separate</i> the dy from the dt so that dy and y are on the same side. If there are t 's in the equation they must go on the same side as dt : $\frac{dy}{f(y)} = g(t)dt$. —	_____
Integrate both sides of the equation (one side with respect to y , the other with respect to t): $\int \frac{1}{f(y)} dy = \int g(t) dt$. —	_____
Solve for the <i>general solution</i> . —	_____
Find the <i>specific solution</i> given the initial condition $y(0) = 1$. —	_____

b. Consider this ODE $y' = \frac{t}{3y^2}$.

Use the shortcut to determine the general solution and find the specific solution given the condition $y(1) = 1$.